Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Python-focused AI Engineer with 3 years of industry experience building and shipping production GenAI applications in a regulated financial services setting: FastAPI microservices in Docker on AWS, RAG pipelines over PostgreSQL with pgvector, tool-calling agents, and a Jenkins and GitLab CI/CD pipeline with LDAP authentication. Worked directly with Compliance, Client Service and Investment stakeholders to turn requirements into working tools; acted as de facto team lead for up to 4 interns and was twice recognised internally with Best Team Project (2024, 2025). Recently completed an MSc in Data Science at the University of Bristol, with a dissertation in partnership with AstraZeneca. Seeking an AI Engineer role building client-facing LLM/GenAI solutions on the Python and data engineering side.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Python Backend, Microservices & Secure Deployment

- Built the backend of internal **GenAI** applications as **Python FastAPI** microservices (uvicorn) in **Docker**, with a **ReactJS** frontend served by **Nginx** on a separate instance; designed the infrastructure on **AWS** using core services such as **EC2, S3, ECR** and **Lambda**, with separate frontend and backend repositories and containers and a one-click **Jenkins** and **GitLab CI/CD** pipeline for dev and prod.
- Implemented **RBAC** for the GenAI apps with a Python authentication function against the company **LDAP** server; restricted access to company private IPs using **AWS Security Groups**; deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the **VAPT** (vulnerability assessment and penetration testing); coordinated with the security team on port whitelisting and development safeguards.

Production GenAI: RAG, Agents & LLM Integration

- Designed and built **RAG** pipelines and knowledge retrieval for internal **GenAI** applications used across Compliance, Client Service and Investment (**LlamaIndex** with **Azure OpenAI GPT-3.5 & Llama-3**), using department-based retrieval: a separate **pgvector** database and **Azure OpenAI** embeddings for each department, so access stayed siloed and similar content from different departments was not confused.
- Built **agentic tool-calling** workflows where AI agents call **Python functions** as tools to run SQL-backed fund-comparison calculations and generate **PDF** comparison reports on user request.
- Built a **Python** email-processing automation that parses over **300 daily emails** from credit rating agencies into structured data, fuzzy-matches them to tracked companies and sends analysts a daily summary, saving **60 manual analyst hours a month**; replaced **AWS Textract** with **Azure OpenAI** when a rating agency began sending emails as images, which also improved accuracy; run as a scheduled job on **AWS SageMaker** via EventBridge and Lambda.
- Refined domain behaviour with **Compliance**, Client Service and Investment staff by releasing to real users and iterating on what broke; guardrailed responses through **system prompts**, versioned every prompt change in **Git** and had the **Compliance** department review each application before release. Allocated models by cost: open-

source **Llama-3 self-hosted on a GPU EC2 instance** for the largest department (Client Service) and stronger **Azure OpenAI** outputs for decision makers, with the ability to switch models.

Data Engineering & Client Delivery

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly (fixed income) and monthly (equity); integrated it with legacy data from analysts' Excel files; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.
- Developed **Power BI** dashboards for fortnightly credit report analysis, migrating financial analysts from Excel to Power BI and improving analyst efficiency by **10 hours a week**; then recreated the dashboards in the **ReactJS** frontend so the wider organisation could use the data without Power BI licensing costs. Used **Python Pandas** across data jobs, backend APIs and RAG pipelines.
- Worked directly with stakeholders in **Client Service, Compliance and Investment** to turn requirements into working tools that replace manual work; acted as **de facto team lead for up to 4 interns** in my final year; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built; I designed the RAG-based AI assistant and worked on the scoring algorithm and data preprocessing for a tool, built as FastAPI microservices with a React and TypeScript frontend, that ranks cancer cell lines by target inclusion and exclusion genes; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). Behaviour steered by prompts; LLM output treated as untrusted: validated with Pydantic, repaired and cached by input hash; rate limits shaped the design (12 calls instead of 108, serialised calls, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Architecture study comparing CNN/ViT/ResNet encoders and LSTM/attention decoders for generating image descriptions, benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **B.Tech Final Year Project:** Udacity self-driving car simulator. (A self-driving car trained using a **Convolution Neural Network (CNN)** model.)

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2, S3, ECR, Lambda, SageMaker), Azure OpenAI, LlamaIndex, Docker, Jenkins, GitLab, Git, Nginx, Linux, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite